Assignment No: 1

Git Basics and Version Control for a Sample Web Application

Aim

To create a GitHub repository for a sample web application and demonstrate basic Git operations: clone, add, commit, push, branch, merge and pull.

Output

1. Clone the repository

```
ronak@ronak:~/devops-task-manager$ git clone https://github.com/ronak/devops-task-manager.git
Cloning into 'devops-task-manager'...
warning: You appear to have cloned an empty repository.
Repository successfully cloned.
ronak@ronak:~/devops-task-manager$ cd devops-task-manager
```

2. Add application files

```
ronak@ronak:~/devops-task-manager$ ls
index.html  script.js  style.css
Student Task Manager application files added.
```

3. Check status

```
ronak@ronak:~/devops-task-manager$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html
        script.js
        style.css
```

4. Stage and commit

```
ronak@ronak:~/devops-task-manager$ git add .
Changes staged successfully.
ronak@ronak:~/devops-task-manager$ git commit -m "Add Student Task Manager application"
[main (root-commit) 8a7c21d] Add Student Task Manager application
 3 files changed.
 create mode 100644 index.html
 create mode 100644 script.js
 create mode 100644 style.css
```

5. Push to GitHub

```
ronak@ronak:~/devops-task-manager$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (5/5), done.
To https://github.com/ronak/devops-task-manager.git
 * [new branch]      main -> main
Application successfully pushed to GitHub.
```

6. Create branch and modify UI

```
ronak@ronak:~/devops-task-manager$ git switch -c feature-ui
Switched to a new branch 'feature-ui'.
User interface modifications completed.
ronak@ronak:~/devops-task-manager$ git add .
ronak@ronak:~/devops-task-manager$ git commit -m "Improve Task Manager user interface"
[feature-ui 4b8e32a] Improve Task Manager user interface
 1 file changed
```

7. Merge into main

```
ronak@ronak:~/devops-task-manager$ git switch main
Switched to branch 'main'.
ronak@ronak:~/devops-task-manager$ git merge feature-ui
Updating 8a7c21d..4b8e32a
Fast-forward
 style.css | 12 ++++++++++++
Branch 'feature-ui' successfully merged.
```

8. Push and pull

```
ronak@ronak:~/devops-task-manager$ git push origin main
Changes successfully pushed to GitHub.
ronak@ronak:~/devops-task-manager$ git pull origin main
Already up to date.
```

9. Commit history

```
ronak@ronak:~/devops-task-manager$ git log --oneline --graph
* 4b8e32a (HEAD -> main, origin/main, feature-ui) Improve Task Manager user interface
* 8a7c21d Add Student Task Manager application
```

Conclusion

Git and GitHub were used to create, clone, commit, branch, merge, push and pull the Student Task Manager application, demonstrating basic version control.